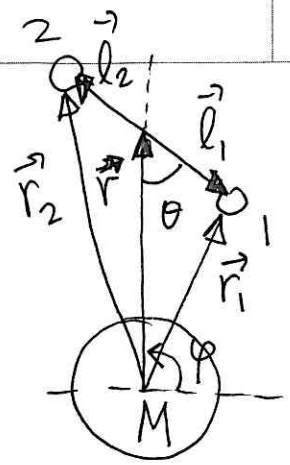
HW06

(P1)



$$\left.\begin{aligned}\vec{r}_1 &= \vec{r} + \vec{l}_1\\ \vec{r}_2 &= \vec{r} + \vec{l}_2\end{aligned}\right\} \qquad \vec{l}_2 = -\vec{l}_1$$

$$\Downarrow$$

$$(\dot{\vec{r}}_1)^2 + (\dot{\vec{r}}_2)^2 = (\dot{\vec{r}} + \dot{\vec{l}}_1)^2 + (\dot{\vec{r}} - \dot{\vec{l}}_1)^2 =$$

$$= 2(\dot{\vec{r}})^2 + 2(\dot{\vec{l}}_1)^2$$

center of mass — relative

(a)

$$\Rightarrow T = \frac{1}{2} m \cdot 2(\dot{\vec{r}})^2 + \frac{1}{2} m \cdot 2 (l\dot{\theta})^2 = m(\dot{r}^2 + r^2\dot{\varphi}^2) + m l^2 \dot{\theta}^2$$

$$U = -\frac{GMm}{r_1} - \frac{GMm}{r_2}$$

$$r_{1,2} = |\vec{r}_{1,2}| = |\vec{r} \pm \vec{l}| = \sqrt{(\vec{r} \pm \vec{l}) \cdot (\vec{r} \pm \vec{l})} =$$

$$= \sqrt{\vec{r}^2 + \vec{l}^2 \pm 2\vec{r}\cdot\vec{l}} = \sqrt{r^2 + l^2 \pm 2rl\cos\theta}$$

$$= r\sqrt{1 + \left(\frac{l}{r}\right)^2 \pm \frac{2l\cos\theta}{r}}$$

Expand in $\frac{l}{r} \ll 1$: to 2nd order

$$\cong r\left\{1 + \frac{1}{2}\left(\frac{l}{r}\right)^2 \pm \frac{1}{2}\frac{2l\cos\theta}{r} - \frac{1}{8}\left(\frac{2l\cos\theta}{r}\right)^2\right\}$$

$$= r\left\{1 \pm \frac{l\cos\theta}{r} + \frac{l^2(1-\cos^2\theta)}{2r^2}\right\}$$

$$\Rightarrow \frac{1}{r_1} + \frac{1}{r_2} = \frac{1}{r}\left\{\frac{1}{1 + \frac{l\cos\theta}{r} + \frac{l^2(1-\cos^2\theta)}{2r^2}} + \frac{1}{1 - \frac{l\cos\theta}{r} + \frac{l^2(1-\cos^2\theta)}{2r^2}}\right\}$$

$$= \frac{1}{r}\left\{1 - \frac{l\cos\theta}{r} - \frac{l^2\sin^2\theta}{2r^2} + \left(\frac{l\cos\theta}{r}\right)^2\right.$$

$$\left. + 1 + \frac{l\cos\theta}{r} - \frac{l^2\sin^2\theta}{2r^2} + \left(\frac{l\cos\theta}{r}\right)^2\right\} =$$

$$= \frac{1}{r}\left\{2 + \frac{l^2}{r^2}\underbrace{\left(2\cos^2\theta - \sin^2\theta\right)}_{3\cos^2\theta - 1}\right\} = \frac{2}{r} + \frac{3l^2}{r^3}\cos^2\theta - \frac{l^2}{r^3}$$

angle-independent correction which we can neglect.

$$\Rightarrow U = -\frac{GM(2m)}{r} - \frac{3GMml^2\cos^2\theta}{r^3}$$

(b) $$L = \underbrace{m\left(\dot{r}^2 + r^2\dot{\varphi}^2\right) + \frac{2GMm}{r}}_{\text{center of mass}} + \underbrace{ml^2\dot{\theta}^2 + \frac{3GMml^2\cos^2\theta}{r^3}}_{\text{relative (small correction to CM)}}$$

Clearly $\frac{d}{dt}\frac{\partial L}{\partial \dot{\varphi}} = 0 \Rightarrow 2mr^2\dot{\varphi} = M_z = \text{const}$

<u>radial</u> $$2m\ddot{r} = 2mr\dot{\varphi}^2 - \frac{2GMm}{r^2}$$

We are told the orbit is circular of Radius R

$$\Rightarrow \ddot{r} = 0 \Rightarrow 2mR\dot{\varphi}^2 - \frac{2GMm}{R^2} = 0$$

but $\varphi = \omega_0 t$ $\qquad \dot{\varphi}^2 = \frac{GM}{R^3} \Rightarrow \omega_0 = \sqrt{\frac{GM}{R^3}}$

orbital angular velocity of satellite.

(c) Now the relative motion:

$$2ml^2\ddot{\theta} = -\frac{\partial}{\partial\theta}\underbrace{\left(-\frac{3GMml^2}{R^3}\cos^2\theta\right)}_{\text{effective potential for } \theta,\ V_{eff}(\theta)}$$

$$\frac{\partial V_{eff}}{\partial\theta} = 0 \Rightarrow \cos\theta\sin\theta = 0 \Rightarrow \theta = 0, \frac{\pi}{2}, \pi.$$

$$V_{eff} = -\frac{3GMml^2}{R^3}\cos^2\theta \Rightarrow \begin{cases} \underline{\theta = 0, \pi} \text{ are } \underline{\text{minimum}} \\ \theta = \frac{\pi}{2} \text{ is maximum} \end{cases}$$

(Note that $\theta_1 = 0$ and $\theta = \pi$ describe the same configuration $\Rightarrow$ need to consider just one case)

(d) $$2ml^2\ddot{\theta} = \frac{3GMml^2}{R^3}(-2\cos\theta\sin\theta)$$

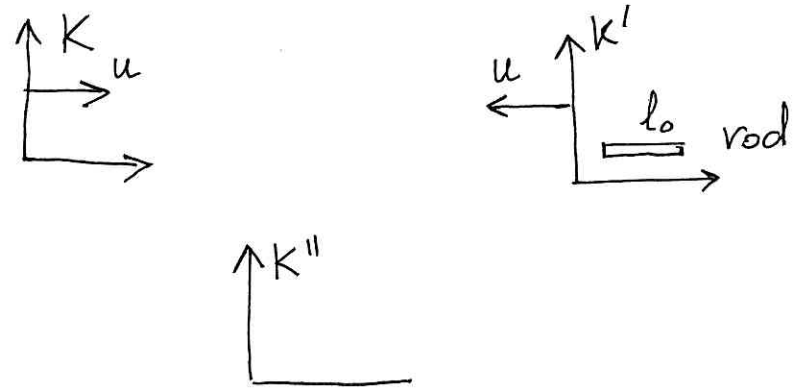
$$\ddot{\theta} = -\frac{3GM}{2R^3}\sin(2\theta)$$

$$\theta = \theta_1 + \theta - \theta_1 = \theta_1 + \delta\theta = \delta\theta$$

$$\delta\ddot{\theta} = -\frac{3GM}{2R^3}\sin(2\delta\theta) = -\frac{3GM}{R^3}\delta\theta$$

$$\Rightarrow \delta\ddot{\theta} = -\Omega^2(\delta\theta) \text{ with } \Omega = \sqrt{\frac{3GM}{R^3}} = \sqrt{3}\,\omega_0.$$

(P2)



In frame K, coordinates of the rod are

$$x = \frac{x'' - u t''}{\sqrt{1 - u^2/c^2}}$$

$$t = \frac{t'' - \frac{u}{c^2} x''}{\sqrt{1 - u^2/c^2}}$$

where x'', t'' are coordinates of the rod in K'' frame.

$$\Rightarrow \quad \frac{dx}{dt} = \frac{dx'' - u\,dt''}{dt'' - \frac{u}{c^2} dx''} = \frac{\frac{dx''}{dt''} - u}{1 - \frac{u}{c^2}\frac{dx''}{dt''}}$$

but $\frac{dx''}{dt''} = -u$ = velocity of the rod in K''.

$\Rightarrow$ its velocity in frame K, $v = \frac{dx}{dt} = \frac{-u-u}{1 - \frac{u}{c^2}(-u)} = \frac{-2u}{1 + \frac{u^2}{c^2}}$

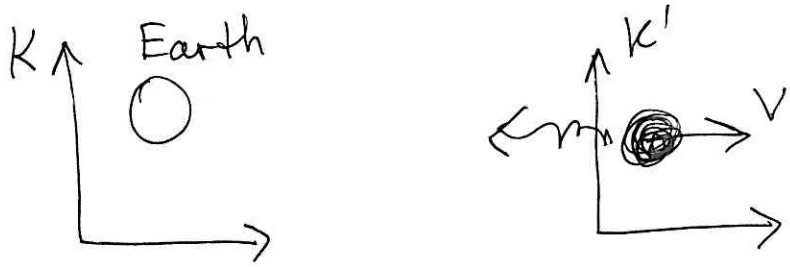
Thus v is the relative velocity between K and K'.

Hence apparence length of rod in K is

$$l = l_0 \sqrt{1 - \frac{v^2}{c^2}} = l_0 \sqrt{1 - \left(\frac{2u/c}{1 + \frac{u^2}{c^2}}\right)^2} = l_0 \frac{\sqrt{\left(1 + \frac{u^2}{c^2}\right)^2 - 4\frac{u^2}{c^2}}}{1 + \frac{u^2}{c^2}}$$

$$= l_0 \frac{1 - \frac{u^2}{c^2}}{1 + \frac{u^2}{c^2}}.$$

(P3)



4-momentum of photon in K':

$$p^i = \left(\frac{E_{em}}{c}, -\frac{E_{em}}{c} \right)$$

because $\vec{p}'_{photon} = \frac{E_{em}}{c}(-\hat{x}')$ in K'

In frame K: $\begin{pmatrix} \frac{E}{c} \\ p \end{pmatrix} = \begin{pmatrix} \gamma & \gamma\beta \\ \gamma\beta & \gamma \end{pmatrix} \begin{pmatrix} \frac{E_{em}}{c} \\ -\frac{E_{em}}{c} \end{pmatrix} = \begin{pmatrix} \gamma(1-\beta)\frac{E_{em}}{c} \\ \gamma(\beta-1)\frac{E_{em}}{c} \end{pmatrix}$

Thus in K too $\vec{p} = p\hat{x} = -\frac{E}{c}$ where

$$\frac{E}{c} = \gamma(1-\beta)\frac{E_{em}}{c} \quad ; \gamma = \frac{1}{\sqrt{1-\beta^2}}, \beta = \frac{V}{c}$$

$$E = \frac{1-\beta}{\sqrt{1-\beta^2}} E_{em} = \sqrt{\frac{1-\beta}{1+\beta}} E_{em}$$

Thus observed energy in K of the photon E is smaller than E_{em} in K'. Since $E = h\nu$, the observed frequency is smaller too — the photon is <u>red-shifted</u> (smaller ν, hence bigger wavelength)

(P4) proper time (in Bert's frame) τ

$$\tau = \int_0^t \sqrt{1 - \frac{v^2}{c^2}}\, dt \qquad \text{where } t = \text{time in Al's frame.}$$

Bert's argument: that Al's situation is symmetric to his and thus Al must too be younger is false because <u>Bert's frame is not inertial</u> (he is making a <u>round-trip</u>).

Hence, there is no symmetry Bert $\Leftrightarrow$ Al.
Hence no paradox.